

Team Phoenix

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Binary Ocular Disease Classification Using ODIR-5K

1. Introduction

In this project, we use the ODIR-5K dataset to develop a **binary deep learning classifier** that distinguishes between healthy and diseased retinal images. Our approach uses transfer learning with EfficientNetV2-S, combined with image preprocessing and augmentation to improve classification performance.

Objectives

- Classify each image into 0 = Normal, 1 = Disease sample (any non-normal).
- Encouraged: A deep-learning pipeline using transfer learning (pretrained ResNet/EfficientNet/DenseNet) for limited labeled medical data.
- Robust preprocessing and training; handle class imbalance (class weighting, weighted sampling, augmentation).
- Patient-level train/validation split to avoid leakage (no same-patient L/R across splits).
- Clinically relevant metrics: sensitivity (recall disease samples), specificity (recall normal samples), F1, confusion matrix, ROCAUC (primary), accuracy alongside.
- Insight into strengths/failures; optional Grad-CAM explainability.

2. Approach

2.1 Dataset & Preprocessing

Keyword Parsing & Ground Truth Extraction

- Used regular expressions (`DISEASE_RE`, `NORMAL_RE`) to parse raw left and right diagnostic text keywords into binary labels (`0: Healthy/Normal`, `1: Defective/Disease`), filtering out uninformative or low-quality images.

- **Why It Was Done:** Standardizes inconsistent free-text clinical diagnostics into a clean target variable.
- **Alternative Approaches:** Relying on raw pre-processed one-hot columns or using a separate NLP model to categorize clinical notes.
- **Model Benefits:** Eliminates mislabeled or poor-quality images prior to training, reducing label noise in the dataset.

Stratified Patient-Level Splitting (Leakage Prevention)

- Aggregated maximum disease labels per `Patient_ID` and executed a 70/15/15 Train/Validation/Test split stratified by patient-level labels.
- **Why It Was Done:** Prevents data leakage where one eye of a patient appears in the training set while the opposite eye appears in validation or testing.
- **Alternative Approaches:** Random image-level splitting across all available fundus files.
- **Model Benefits:** Ensures true out-of-sample evaluation and prevents artificially inflated validation metrics.

Fundus Preprocessing, CLAHE, and Parallel Caching

- Auto-cropped black background borders (`crop_fundus`), converted images to LAB color space to apply Contrast Limited Adaptive Histogram Equalization (`CLAHE`) on the L-channel, resized to 384 X 384, and cached results to disk using `ThreadPoolExecutor`.
- **Why It Was Done:** Standardizes aspect ratios, enhances vascular/lesion contrast, and pre-renders images to prevent GPU starvation.
- **Alternative Approaches:** Dynamic on-the-fly preprocessing during data loading or simple bilinear resizing without cropping/CLAHE.
- **Model Benefits:** Removes CPU bottlenecks during training and amplifies micro-lesion visibility.

Dynamic Input Pipeline & Data Augmentations

- Constructed a tf.data pipeline applying dynamic horizontal/vertical flips, 90 degree rotations, random crop/zoom (80- 100%), and brightness/contrast adjustments (Note: Corrects reference to `ConvNeXt` and manual left-to-right eye conversions).

- **Why It Was Done:** Exposes the network to structural variations, preventing overfitting on high-resolution fundus patterns.
- **Alternative Approaches:** Offline static dataset expansion or basic Keras ImageDataGenerator.
- **Model Benefits:** Runs GPU-accelerated augmentations natively within the TensorFlow graph while prefetching the next batch.

2.2 Model Architecture & Training

Environment Setup & Distributed Strategy

- Configured deterministic random seeds (`SEED = 42`), enabled GPU memory growth, set global `mixed_float16` precision, and initialized TensorFlow's `MirroredStrategy`.
- **Why It Was Done:** Memory growth prevents VRAM allocation crashes, mixed-precision speeds up GPU tensor calculations, and `MirroredStrategy` enables multi-GPU scalability.
- **Alternative Approaches:** Running single-GPU execution under standard `float32` precision without memory growth constraints.
- **Model Benefits:** Reduces GPU memory usage by up to 50%, allowing higher resolution inputs (384 X 384) and larger batch sizes without out-of-memory (OOM) errors.
- Configured deterministic random seeds (`SEED = 42`), enabled GPU memory growth, set global `mixed_float16` precision, and initialized TensorFlow's `MirroredStrategy` across two Kaggle T4 GPUs.

Model Architecture & Transfer Learning Backbone

- Utilized transfer learning with an `EfficientNetV2S` backbone (falling back to `EfficientNetB4`), attached to a custom head: Global Average Pooling, Dropout (0.4/0.3), Dense(256, ReLU), and a 1-unit Sigmoidal classifier with label smoothing (0.08)
- **Why It Was Done:** EfficientNet backbones offer strong feature extraction with high parameter efficiency.
- **Alternative Approaches:** ResNet50, DenseNet, Vision Transformers (ViT), or training a model from scratch.
- **Model Benefits:** Leverages ImageNet features for visual representation while label smoothing prevents overly confident output probabilities.

Imbalance Handling & Two-Phase Training Strategy

- Computed class weights for loss balance. Phase 1 (4 epochs) trained the classification head with the backbone frozen using **AdamW** ($\text{lr}=1\text{e-}3$). Phase 2 (18 epochs) unfroze the top 80% of backbone layers—keeping Batch Normalization layers frozen—using a **CosineDecay** learning rate schedule ($5\text{e-}5$ to minimum).
- **Why It Was Done:** Prevents randomly initialized top layers from corrupting pre-trained backbone weights and stabilizes Batch Normalization statistics during fine-tuning.
- **Alternative Approaches:** End-to-end training in a single phase or unfreezing Batch Normalization layers.
- **Model Benefits:** Mitigates class imbalance bias and yields smooth convergence during deep feature fine-tuning.

2.3 Inference & Evaluation

Test-Time Augmentation (TTA) & Decision Threshold Tuning

- Implemented vectorized 3-view TTA (original, horizontal flip, vertical flip) on validation/test sets and scanned classification thresholds (0.20-0.80) on validation predictions to maximize accuracy.
- **Why It Was Done:** Averages prediction variance across orientations and optimizes decision boundaries for imbalanced datasets.
- **Alternative Approaches:** Single-pass evaluation using a default 0.50 decision threshold.
- **Model Benefits:** Improves classification stability on borderline samples and optimizes trade-offs between sensitivity and specificity.

Clinical Performance Evaluation & Metrics

- Evaluated the test set using primary metric ROC-AUC, alongside Accuracy, F1-Score, Sensitivity (Recall), Specificity, and full Confusion Matrix output.
- **Why It Was Done:** Clinical deployment requires evaluation beyond simple accuracy, specifically tracking false negative rates (Sensitivity) and false positive rates (Specificity).
- **Alternative Approaches:** Evaluating solely on loss or raw accuracy.

- **Model Benefits:** Delivers a complete picture of diagnostic reliability across both healthy and defective retinal categories.

3. Results

Metrics Evaluated: Primary performance benchmarked via ROC-AUC alongside Accuracy, F1-Score, Sensitivity (Recall), Specificity, and full Confusion Matrix analysis.

ROC-AUC : 85.01%

Accuracy: 78.53%

F1 Score: 78.20%

Sensitivity (Recall): 70.78%

Specificity: 87.78%

Confusion Matrix :

	Predicted healthy	Predicted defective
Actual healthy	388	54
Actual defective	154	373

Optimization Impact: Dynamic thresholding paired with TTA stabilized output probabilities, significantly reducing false negatives on minority disease classes compared to standard 0.50 decision boundaries.

4. Limitations

- **Resolution Loss:** Downsampling high-resolution fundus scans to 384x384 pixels risks obscuring extremely subtle micro-lesions (such as microaneurysms).
- **Rule-Based Label Extraction:** Ground-truth reliance on regex keyword matching from free-text reports can misclassify nuanced or complex diagnostic descriptions.
- **Compute Constraints:** Multi-pass TTA and deep fine-tuning require dedicated GPU memory growth and mixed-precision acceleration.

5. Failure Cases

- **Low-Quality Artefacts:** Scans severely degraded by cataracts, lens glare, or extreme underexposure, causing CLAHE to amplify background image noise rather than vascular structures.
- **Atypical Pathologies:** Borderline cases featuring overlapping co-occurring eye conditions or non-standard lesion patterns that fall outside the learned feature representations.

6. Bias & Generalizability

- **Dataset Bias:** Class distribution imbalance was addressed via weighted loss functions, but underlying demographic or hardware-specific biases from the source Kaggle dataset remain present.
- **Generalizability Scope:** Patient-level stratification ensures robust out-of-sample validation within this dataset, but real-world clinical deployment across different hospital camera vendors (domain shift) requires external validation.

7. Future Work

- Future developments will focus on expanding the model's clinical versatility and addressing resolution constraints. These include integrating an automated image-quality filter to detect and reject severely degraded or glare-affected scans before inference, expanding from binary classification to multi-label diagnosis for co-occurring eye conditions, and evaluating the model on external multi-center datasets from diverse camera manufacturers to verify real-world domain generalizability.
- We can also train the model to look at smaller, zoomed-in pieces of the image instead of the whole file at once. This helps it spot tiny disease details easily without overloading the computer's memory.

8. Conclusion

- This project establishes a reliable, leak-free medical imaging framework for automated retinal pathology classification. By combining targeted CLAHE contrast enhancement and patient-stratified splitting with a two-phase EfficientNetV2S fine-tuning strategy, the pipeline balances strong feature extraction with efficient GPU resource usage. Coupled with test-time augmentation and dynamic decision thresholding to minimize critical false negatives, the system delivers a robust, scalable foundation for automated ophthalmic screening.

Screenshots Of Output

```
Epoch 5: finished saving model to /kaggle/working/checkpoints/best_fundus_90.keras
279/279 ————— 102s 363ms/step - auc: 0.9176 - loss: 0.0483 - val_auc: 0.9165 - val_loss: 0.0502
Epoch 6/18
279/279 ————— 0s 327ms/step - auc: 0.9211 - loss: 0.0445
Epoch 6: val_auc improved from 0.91649 to 0.91707, saving model to /kaggle/working/checkpoints/best_fundus_90.keras

Epoch 6: finished saving model to /kaggle/working/checkpoints/best_fundus_90.keras
279/279 ————— 102s 364ms/step - auc: 0.9303 - loss: 0.0451 - val_auc: 0.9171 - val_loss: 0.0477
Epoch 7/18
279/279 ————— 0s 326ms/step - auc: 0.9384 - loss: 0.0416
Epoch 7: val_auc did not improve from 0.91707
279/279 ————— 99s 355ms/step - auc: 0.9414 - loss: 0.0424 - val_auc: 0.9167 - val_loss: 0.0485
Epoch 8/18
279/279 ————— 0s 328ms/step - auc: 0.9447 - loss: 0.0397
Epoch 8: val_auc did not improve from 0.91707
279/279 ————— 100s 356ms/step - auc: 0.9521 - loss: 0.0389 - val_auc: 0.9164 - val_loss: 0.0494
Epoch 9/18
279/279 ————— 0s 328ms/step - auc: 0.9566 - loss: 0.0347
Epoch 9: val_auc did not improve from 0.91707
279/279 ————— 100s 357ms/step - auc: 0.9594 - loss: 0.0359 - val_auc: 0.9156 - val_loss: 0.0507
Epoch 10/18
279/279 ————— 0s 328ms/step - auc: 0.9595 - loss: 0.0329
Epoch 10: val_auc did not improve from 0.91707
279/279 ————— 100s 357ms/step - auc: 0.9677 - loss: 0.0327 - val_auc: 0.9120 - val_loss: 0.0544
Epoch 11/18
279/279 ————— 0s 327ms/step - auc: 0.9694 - loss: 0.0297
Epoch 11: val_auc improved from 0.91707 to 0.91714, saving model to /kaggle/working/checkpoints/best_fundus_90.keras

Epoch 11: finished saving model to /kaggle/working/checkpoints/best_fundus_90.keras
279/279 ————— 102s 363ms/step - auc: 0.9738 - loss: 0.0296 - val_auc: 0.9171 - val_loss: 0.0537
```

FINAL PERFORMANCE METRICS (Competition Binary Task)

```
ROC-AUC (Primary):      0.8657
Accuracy:               0.7843 (78.4%)
F1 Score:               0.7961
Sensitivity (Recall):   0.7742
Specificity:            0.7964
```

CONFUSION MATRIX:

	Predicted Healthy (0)	Predicted Defective (1)
Actual Healthy (0)	352	90
Actual Defect. (1)	119	408

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